

Name: _____

Date: _____

PROBABILITY, (TREE DIAGRAMS)

LO: I can use tree diagrams to find probabilities of combined events.

Tree diagrams

A **tree diagram** shows all possible outcomes of two or more events. Each **branch** represents one outcome and is labelled with its probability.

Rules: Multiply along branches (AND). Add between paths (OR). Branches from one point always sum to 1.

Quick examples

●	$P(A \text{ and } B) = P(A) \times P(B)$	multiply along branches
●	$P(A \text{ or } B) = \text{add separate paths}$	add outcomes
●	Branches from a node sum to 1	all possibilities covered
●	$P(A \text{ and } B) = P(A) + P(B)$	should multiply for AND

Key idea

Multiply along branches for combined events. Add between separate paths for 'or' questions.

$$P(A \text{ then } B) = P(A) \times P(B)$$

AND = multiply; OR = add

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - two coins

A coin is flipped twice. Find $P(\text{two heads})$.

$$\begin{aligned} P(H) &= \frac{1}{2} \text{ for each flip} \\ P(HH) &= \frac{1}{2} \times \frac{1}{2} \\ &= \frac{1}{4} \end{aligned}$$

$$P(\text{two heads}) = \frac{1}{4}$$

Multiply along the HH branch.

Example 2 - with replacement

A bag has 3 red, 2 blue counters. One is drawn, replaced, then another drawn. Find $P(\text{both red})$.

$$\begin{aligned} P(\text{red}) &= \frac{3}{5} \text{ each time (replaced)} \\ P(RR) &= \frac{3}{5} \times \frac{3}{5} \\ &= \frac{9}{25} \end{aligned}$$

$$P(\text{both red}) = \frac{9}{25}$$

With replacement, probabilities stay the same.

Example 3 - without replacement

Same bag (3R, 2B). Without replacement, find $P(\text{both red})$.

$$\begin{aligned} \text{First pick: } P(R) &= \frac{3}{5} \\ \text{Second pick (one red gone): } P(R) &= \frac{2}{4} = \frac{1}{2} \\ P(RR) &= \frac{3}{5} \times \frac{1}{2} = \frac{3}{10} \end{aligned}$$

$$P(\text{both red}) = \frac{3}{10}$$

Without replacement, the second probability changes.

Example 4 - at least one

Two coins are flipped. Find $P(\text{at least one head})$.

$$\begin{aligned} P(\text{no heads}) &= P(TT) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4} \\ P(\text{at least one H}) &= 1 - P(TT) \\ &= 1 - \frac{1}{4} = \frac{3}{4} \end{aligned}$$

$$P(\text{at least one head}) = \frac{3}{4}$$

Use the complement: $1 - P(\text{none})$.

YOUR TURN – PRACTICE (deliberate practice)

1. Independent events

A coin is flipped and a die is rolled. Find each probability.

- a) P(head and 6)
- b) P(tail and even)
- c) P(head and prime)
- d) P(tail and greater than 4)

2. With replacement

Bag: 4 red, 6 blue. Draw one, replace, draw another.

- a) P(red, red)
- b) P(blue, blue)
- c) P(red, blue)
- d) P(same colour)

3. Without replacement

Bag: 5 red, 3 blue. Draw two without replacement.

- a) P(red, red)
- b) P(blue, blue)
- c) P(red then blue)
- d) P(different colours)

4. At least one

Use $P(\text{at least one}) = 1 - P(\text{none})$.

- a) Two coins: P(at least one tail)
- b) Two dice: P(at least one 6)
- c) P(rain) = 0.3 each day, 2 days: P(at least one rainy day)
- d) Three coins: P(at least one head)

5. Mixed tree diagram problems

Solve each problem using a tree diagram.

- a) P(win) = 0.6. Two games. P(win both)
- b) Bag: 2R, 3B. Two draws without replacement. P(one of each)
- c) P(bus late) = 0.2. Mon and Tue. P(at least late once)
- d) Spinner: P(red) = $\frac{1}{3}$. Spun twice. P(not red both times)
- e) Coin and spinner (1,2,3,4). P(head and even)

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) For combined events, multiply along branches ☐

Correct answer: _____
Reason: _____

b) $P(A \text{ and } B) = P(A) + P(B)$ ☐

Correct answer: _____
Reason: _____

c) Without replacement, the probabilities change for the second pick ☐

Correct answer: _____
Reason: _____

d) Branches from one node always add to more than 1 ☐

Correct answer: _____
Reason: _____

e) $P(\text{at least one}) = 1 - P(\text{none})$ ☐

Correct answer: _____
Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A bag contains **4 red and 6 blue** balls. Two balls are drawn without replacement. Draw a complete tree diagram and find the probability of getting **exactly one red** ball.

Expression: _____

Simplified: _____

b) The probability that a bus is late is **0.15**. On three consecutive days, find the probability that the bus is late on **exactly two** of the three days.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

- Have I shown all my working clearly?
- Did I check my answer using a different method?
- Does my answer look reasonable?

